

Harry Nguyen

+1 470-667-9000 | harrynguyenswe@gmail.com | linkedin.com/in/harrynguyen26 | github.com/HarryNguyen2662 | timoto.ai

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec** — selected Protobuf over custom serialization for schema evolution and zero-copy cross-process message passing
- Profiled settings navigation as p99 bottleneck at 1,200ms; designed **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Architected event-driven **TypeScript/React** observer system decoupling UI state propagation across **25K+ lines** of Chromium at **95% test coverage**; reviewed and adopted into production branch by senior Chrome engineers

CoderPush | Software Engineer Intern

Sep 2023 – Dec 2023

- Diagnosed **DynamoDB hot-partition failure** in high-throughput data pipeline at **9,000+ req/sec**; redesigned partition key strategy and access patterns — **30% read throughput improvement** under sustained concurrency
- Designed idempotent **REST APIs** with **Redis** distributed caching (**85% cache hit rate**) and exponential backoff — enforced **exactly-once semantics** preventing duplicate transactions under network partitions

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on **AWS** with JWT-based auth, auto-scaling supporting **500+ concurrent users**; diagnosed N+1 query bottleneck, redesigned **PostgreSQL** indexing achieving **sub-100ms** for 10,000+ records

PERSONAL PROJECTS

TiMoto AI – Founding Engineer, Real-Time Data Platform | Go, Python, gRPC, Django, AWS, Terraform, Kubernetes

- Resolved **production deadlock under concurrent gRPC calls** as **sole engineer** — traced root cause via structured logging to circular resource acquisition; redesigned call sequencing with explicit lock ordering — **100% success rate** at **sub-50ms p99**
- Enforced **exactly-once semantics** under concurrent session load — circuit breaker pattern with health-check-triggered auto-rollback; zero duplicate processing at production traffic
- Selected **vLLM** with **PagedAttention** for zero OOM failures — traded virtual memory overhead for disk-backed KV cache paging; continuous batching manages throughput without page faults under load
- Architected multi-AZ **ECS Fargate** with **Terraform IaC** — **99.9% uptime, 44% cost reduction** to \$40-60/month; owned infra, backend, gRPC layer, CI/CD, and monitoring as sole engineer

Pulumi – Open Source Contributor | Go, TypeScript, IaC

24.4K★ github.com/pulumi/pulumi

- Contributed **Go** CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness proofs under concurrent operations and partial failures

TECHNICAL SKILLS

Languages: TypeScript, Go, Python, C++, Java, JavaScript, Rust, SQL, Bash

Distributed Systems & Data Infrastructure: gRPC, Protocol Buffers, Kafka, Exactly-once semantics, Circuit breakers, Raft/Paxos, Linearizability, Fault tolerance

Performance & Concurrency: Lock-free data structures, p99 profiling, Mutex contention analysis, Memory-aware scheduling, Hash-based partitioning

Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS, DynamoDB), GCP, Kubernetes, Helm, Docker, Terraform, CI/CD, GitHub Actions

Databases & Caching: PostgreSQL, Redis, Elasticsearch, DynamoDB